

7.2 Trigonometric Integrals

Example 1

Suppose we wish to evaluate $\int \cos^2(x) \sin^3(x) \, dx$. How might we approach this?

Can we transform the integrand into a form that has only one sine factor and the remainder of the expression involves only cosine? Hint: trig identity

Why is this helpful?

What if the power of cosine was odd?

In general, given a trig integral with powers of sine and/or cosine, if one power is odd, use the **Pythagorean trigonometric identity**:

- $\sin^2(\theta) =$

- $\cos^2(\theta) =$

Example 2

Evaluate $\int \cos^5(x) \sin^2(x) \, dx$

Example 3

Evaluate $\int \sin^2(x) \cos^2(x) \, dx$.

Does the Pythagorean identity help with this example? What other trig identities do you know involving sine and cosine?

Summary: Evaluating $\int \sin^m(x) \cos^n(x) dx$

Case 1: m or n is odd

- Choose the trig function with the **odd power** (if both are odd, just choose one!).
- Save one factor to be used for du .
- Convert the remaining **even power** using the appropriate form of the identity:

$$\sin^2(x) + \cos^2(x) = 1 \quad \Rightarrow \quad \begin{cases} \sin^2(x) = 1 - \cos^2(x) \\ \cos^2(x) = 1 - \sin^2(x) \end{cases}$$

- Apply u -substitution.

Case 2: m and n are BOTH even

- Reduce **all even powers** using the identities

$$\begin{aligned} \sin^2(\theta) &= \frac{1 - \cos(2\theta)}{2} \\ \cos^2(\theta) &= \frac{1 + \cos(2\theta)}{2} \end{aligned}$$

- You may need to expand and apply the previous step multiple times until you have eliminated all even powers.
- It is sometimes helpful to use the identity $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$.

Extra Examples

Try the following examples on your own before watching the videos!

1. $\int \cos^4(x) \, dx$

2. $\int_{\pi/3}^{\pi/2} \frac{\sin^2(x)}{\sqrt{1 - \cos(2x)}} dx$

How can we use the Pythagorean trigonometric identity to relate $\sec(x)$ and $\tan(x)$? $\csc(x)$ and $\cot(x)$?

Trig Identities

- $\tan(x) =$

- $\tan^2(x) =$

- $\sec(x) =$

- $\sec^2(x) =$

- $\cot(x) =$

- $\cot^2(x) =$

- $\csc(x) =$

- $\csc^2(x) =$

How will integrals involving powers of secant and tangent be different than those involving sine and cosine?

Example 4

Evaluate $\int \tan^3(x) \sec(x) \, dx$.

Example 5

Evaluate $\int \cot^2(x) \csc^4(x) \, dx$.

Summary: Evaluating $\int \tan^m(x) \sec^n(x) dx$

Case 1: n is even (i.e. power of secant is even)

- Save one factor of $\sec^2(x)$ to be used for du .
- Convert the remaining **even powers** of secant using $\sec^2(x) = 1 + \tan^2(x)$
- You should be left with an integrand involving only tangent, multiplied by $\sec^2(x)$.
- Apply u -substitution ($u = \tan(x)$).

Case 2: m is odd (i.e. power of tangent is odd)

- Save one factor of $\sec(x) \tan(x)$ to be used for du .
- Convert the remaining **even powers** of tangent using $\tan^2(x) = \sec^2(x) - 1$
- You should be left with an integrand involving only secant, multiplied by $\sec(x) \tan(x)$.
- Apply u -substitution ($u = \sec(x)$).

NOTE: You may need two basic integrals:

- $\int \tan(u) du = \ln |\sec(u)| + C$
- $\int \sec(u) du = \ln |\sec(u) + \tan(u)| + C$

Sometimes, the cases are not clear cut. You may need to use integration by parts, or modify the integrand using another identity.

Example 6

Evaluate $\int \sec^3(x) \, dx$.

Extra example

Try this one your own before watching the video!

Evaluate $\int \tan^4(x) \, dx$.